

BUSINESS PROPOSAL DOCUMENT · MULIT-AGENTS BUSINESS INTELLIGENCE

Problem Statement, Agents Definition and Proposed Success Critiria

Secure, AI-Powered Multilingual Multi-Agent Assistant for SME Business Performance Monitoring.

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Team	Yiting · Sonne · Tiffany · Malar
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1. Executive Summary

Businesses often have access to large amounts of sales, customer, financial, and inventory data, yet answering a seemingly simple question such as “**Why did my sales decrease?**” can require significant manual analysis across multiple systems and departments.

We propose **Queen Bee**, an AI-powered multi-agent business intelligence platform designed to transform complex business questions into evidence-based insights and actionable recommendations.

Queen Bee acts as the central orchestration agent. When a business owner asks a question, Queen Bee identifies the underlying business problem, determines which specialist agents are required, coordinates their analysis, correlates their findings, and presents the resulting evidence and recommendations through a business dashboard.

The proposed ecosystem consists of:

- **Queen Bee** — Orchestrates the investigation and determines which specialists are required.
- **Sales Bee** — Analyses revenue, sales performance, products, channels, and margins.
- **Customer Bee** — Analyses enquiries, leads, conversion, customer behaviour, and acquisition.
- **Finance / Accounting Bee** — Analyses financial indicators, costs, profitability, cash flow, and accounting information.
- **Inventory Bee** — Analyses stock levels, product availability, stock-outs, and inventory-related sales constraints.
- **Advisor Bee** — Correlates findings from the specialist agents and develops an overall business explanation and recommended actions.
- **Guard Bee** — Validates proposed actions against business rules, permissions, approval requirements, and risk controls.
- **Audit Bee** — Maintains a transparent execution record of agent activities, evidence, decisions, and actions.
- **Business Dashboard** — Presents evidence, findings, recommendations, approvals, and a visual execution trace.

The objective is to provide business owners with a single intelligent interface for understanding **what happened, why it happened, what can be done, and what evidence supports the recommendation.**

2. Business Problem

Business owners frequently ask questions such as:

- Why did sales decrease?
- Which products are underperforming?

- Why are leads not converting?
- Are stock shortages affecting revenue?
- Why did profit decrease even though sales increased?
- Which customers are becoming inactive?
- Which products or channels have declining margins?
- What actions should the business take?

Answering these questions typically requires information from multiple business functions.

For example, a decline in sales may be caused by:

1. Reduced customer enquiries.
2. Lower lead-to-sale conversion.
3. Product availability issues.
4. Changes in product mix.
5. Reduced margins.
6. Changes in pricing.
7. Increased competition or changes in customer behaviour.
8. Financial or operational constraints.

Looking at sales data alone may therefore produce an incomplete explanation.

The proposed platform addresses this problem by enabling multiple specialised AI agents to investigate different aspects of the same business question and then combining their findings into a unified analysis.

3. Proposed Solution

Queen Bee Multi-Agent Architecture

The platform uses a **multi-agent architecture** in which each agent has a defined area of responsibility.

The business owner interacts primarily with Queen Bee rather than needing to understand the underlying data sources or analytical processes.

Example

Business Owner:

“Why did my sales decrease last month?”

Investigation workflow

Business Owner → Queen Bee → Specialist Bees → Advisor Bee → Guard Bee + Audit Bee → Dashboard

Queen Bee interprets the question and determines that the investigation requires analysis from:

- Sales Bee
- Customer Bee
- Finance / Accounting Bee
- Inventory Bee

Each specialist independently investigates its area and produces findings supported by relevant data.

Advisor Bee then correlates these findings to determine how the different factors may be connected.

Before any recommended action is executed, Guard Bee validates the recommendation against predefined controls and approval requirements.

Audit Bee records the entire process, creating an auditable execution trail.

The dashboard then presents the business owner with the evidence, explanation, recommendations, and execution trace.

4. Agent Responsibilities

4.1 Queen Bee — Orchestration Agent

Queen Bee is the central intelligence and workflow orchestration layer.

Its responsibilities include:

- Understanding the business owner's question.
- Identifying the underlying business objective.
- Determining which specialist agents are relevant.
- Creating the investigation workflow.
- Providing context and relevant parameters to specialist agents.
- Coordinating agent execution.
- Managing dependencies between investigations.
- Detecting missing information or conflicting findings.
- Escalating issues where human input is required.

Queen Bee does not attempt to perform every analysis itself. Instead, it delegates work to specialised agents.

4.2 Sales Bee — Revenue & Margin Analysis

Sales Bee focuses on commercial performance.

It can analyse:

- Revenue trends.
- Sales volume.
- Average order value.
- Product performance.
- Product mix.
- Sales channels.
- Geographic performance.
- Pricing changes.
- Gross margin.
- Revenue versus previous periods.
- Sales targets and variance.

Example finding:

“Revenue decreased 12% compared with the previous month, primarily driven by a decline in Product Category A and a reduction in average order value.”

4.3 Customer Bee — Enquiries & Leads

Customer Bee analyses the customer acquisition and conversion funnel.

It can investigate:

- Customer enquiries.
- Leads generated.
- Lead sources.
- Lead conversion rates.
- Customer acquisition trends.
- Repeat customers.
- Customer retention.

- Customer inactivity.
- Conversion by sales channel.
- Changes in customer behaviour.

Example finding:

“Customer enquiries remained stable, but lead-to-order conversion decreased from 18% to 12%.”

This helps distinguish between a **demand problem** and a **conversion problem**.

4.4 Finance / Accounting Bee — Financial Indicators

Finance / Accounting Bee analyses financial performance and accounting indicators.

It can examine:

- Revenue.
- Cost of goods sold.
- Operating expenses.
- Gross profit.
- Net profit.
- Cash flow indicators.
- Accounts receivable.
- Accounts payable.
- Budget versus actual performance.
- Financial ratios.
- Cost changes.

Example finding:

“Although revenue declined 8%, gross margin declined by 15% due to increased supplier costs in two major product categories.”

4.5 Inventory Bee — Stock & Availability

Inventory Bee investigates whether operational stock constraints are affecting business performance.

It can analyse:

- Current stock levels.

- Stock-outs.
- Product availability.
- Inventory turnover.
- Slow-moving products.
- Fast-moving products.
- Reorder levels.
- Supplier lead times.
- Backorders.
- Stock availability by location.

Example finding:

“Three of the top ten revenue-generating products experienced stock-outs during 22% of the reporting period.”

This allows the platform to connect inventory conditions with sales performance.

5. Advisor Bee — Cross-Functional Business Reasoning

Advisor Bee is responsible for synthesising the specialist findings.

Rather than simply displaying four independent reports, Advisor Bee establishes relationships between them.

For example:

Inventory Bee:

Product A was unavailable for 9 days.

Sales Bee:

Product A normally contributes 18% of monthly revenue.

Customer Bee:

Enquiries for Product A remained high.

Finance Bee:

Product A has above-average gross margin.

Advisor Bee can correlate these findings and present a consolidated explanation:

“The sales decline appears to be significantly associated with the availability of Product A. Demand remained strong, but repeated stock-outs constrained sales of a high-revenue, high-margin product.”

Importantly, the system should distinguish between:

- **Observed facts**
- **Calculated indicators**
- **Agent interpretations**
- **Recommendations**
- **Actions requiring approval**

This separation improves transparency and reduces the risk of treating an AI-generated interpretation as an established fact.

6. Guard Bee — Governance & Action Validation

Guard Bee provides a governance and control layer between recommendations and execution.

Its responsibilities include:

- Validating proposed actions.
- Checking business rules.
- Checking user permissions.
- Identifying actions requiring approval.
- Applying spending or operational limits.
- Preventing unauthorised actions.
- Validating required parameters.
- Escalating sensitive actions to human approval.

For example, Advisor Bee may recommend:

“Increase the reorder quantity for Product A.”

Guard Bee determines whether the proposed action:

- Falls within predefined inventory rules.
- Is within purchasing limits.
- Requires manager approval.
- Meets supplier constraints.

- Can be automatically executed.

The objective is to ensure that AI-generated recommendations do not automatically become business actions without appropriate controls.

7. Audit Bee — Transparency & Auditability

Audit Bee maintains a complete record of the investigation.

The audit trail can include:

- Original business question.
- Timestamp.
- Agents invoked.
- Data sources accessed.
- Queries or analytical tasks performed.
- Findings generated.
- Evidence supporting findings.
- Recommendations produced.
- Guard Bee validation results.
- Human approvals.
- Actions executed.
- Execution results.
- Exceptions or errors.

This creates a **visual and auditable chain of reasoning and execution** without requiring the business owner to inspect the underlying technical processes.

8. Business Dashboard

The dashboard provides a single interface through which business owners can understand the investigation.

A typical investigation could be displayed as:

Question

Why did sales decrease?

Key finding

Revenue decreased 12% compared with the previous month.

Contributing factors

Area	Finding
Sales	Product A revenue decreased significantly
	Customer Enquiries remained relatively stable
Inventory	Product A experienced repeated stock-outs
Finance	Product A has above-average margin

Recommended actions

1. Review replenishment levels for Product A.
2. Investigate supplier lead-time issues.
3. Review sales conversion performance.
4. Monitor Product A availability and revenue over the next reporting period.

Evidence

The dashboard should allow the user to drill down into the underlying evidence supporting each finding.

Execution Trace

A visual workflow can show:

Question → Queen Bee → Sales Bee / Customer Bee / Finance Bee / Inventory Bee → Advisor Bee → Guard Bee → Approval → Action → Audit Bee

This gives the business owner visibility into **how the answer was produced**.

9. Proposed End-to-End Workflow

Step 1 — Business Question

The business owner asks a natural-language question.

“Why did my sales decrease?”

Step 2 — Question Interpretation

Queen Bee interprets the question and identifies the required analytical domains.

Step 3 — Specialist Investigation

Relevant specialist agents independently analyse their respective domains.

Step 4 — Evidence Collection

Each agent returns structured findings with supporting evidence and confidence/context information.

Step 5 — Cross-Functional Correlation

Advisor Bee compares the findings and identifies relationships between business factors.

Step 6 — Recommendation

Advisor Bee proposes potential actions based on the evidence.

Step 7 — Governance Validation

Guard Bee validates recommendations against business policies, permissions and approval requirements.

Step 8 — Human Approval

Where required, the business owner or authorised employee approves the proposed action.

Step 9 — Execution

Approved actions can be executed through connected business systems.

Step 10 — Audit

Audit Bee records the investigation, recommendation, approval and execution history.

Step 11 — Dashboard

The final result is presented through the dashboard with:

- Business explanation.
- Supporting evidence.
- Recommendations.
- Approval status.
- Actions taken.
- Execution trace.
- Audit history.

10. Key Differentiators

The proposed platform is differentiated by combining **multi-agent intelligence, cross-functional analysis, governance and auditability**.

10.1 Natural-Language Business Interaction

Business owners can ask questions using ordinary business language rather than learning complex reporting tools.

10.2 Specialist AI Agents

Each business domain has a dedicated specialist responsible for a clearly defined analytical function.

10.3 Cross-Functional Reasoning

The platform does not analyse sales, customers, finance and inventory in isolation. Advisor Bee connects the findings.

10.4 Governed AI Actions

Recommendations pass through Guard Bee before execution, allowing businesses to establish appropriate controls.

10.5 Full Audit Trail

Audit Bee provides traceability from the original question through to recommendations and actions.

10.6 Explainable Business Insights

The system focuses not only on providing an answer, but also on showing **why that answer was produced** and what evidence supports it.

11. Example Business Scenario

A business owner notices that monthly sales have fallen.

They ask:

“Why did my sales decrease?”

Queen Bee determines that Sales, Customer, Finance and Inventory analysis is required.

Sales Bee identifies a 12% revenue decline.

Customer Bee finds that enquiries have remained relatively stable, but conversion has fallen.

Inventory Bee discovers that several high-performing products experienced stock shortages.

Finance Bee identifies increasing supplier costs and declining margins.

Advisor Bee correlates the findings and identifies two major areas requiring attention:

1. Product availability.
2. Declining conversion and margin performance.

The dashboard presents the evidence and proposed actions.

Guard Bee then validates whether the proposed inventory and pricing actions can be executed automatically or require management approval.

Once approved, the relevant action is executed and Audit Bee records the complete process.

The business owner therefore moves from:

“Something is wrong with sales.”

to:

“Here is the evidence explaining the decline, the likely contributing factors, the recommended actions, the approvals required, and the complete record of how the recommendation was generated.”

12. Proposed Technology Concept

The platform can be designed around a modular architecture consisting of:

- **AI Agent Orchestration Layer**
- **Specialist Agent Layer**
- **Business Data Integration Layer**
- **Knowledge / Context Layer**
- **Rules & Governance Layer**
- **Action / Workflow Layer**
- **Audit & Observability Layer**
- **Business Intelligence Dashboard**

Potential data sources could include:

- Accounting systems.
- ERP systems.
- CRM systems.
- Point-of-sale systems.
- E-commerce platforms.
- Inventory management systems.
- Customer support systems.
- Spreadsheets and business databases.

The architecture should use structured interfaces between agents so that each agent produces consistent outputs such as:

Finding → Evidence → Business Impact → Confidence / Qualification → Recommendation → Required Approval

This provides a foundation for scalable agent collaboration.

13. Implementation Roadmap

Phase 1 — Foundation

- Establish data integration architecture.
- Define common business data model.
- Implement Queen Bee orchestration.
- Implement Sales Bee.
- Develop initial dashboard.
- Establish audit logging.

Phase 2 — Multi-Domain Intelligence

- Implement Customer Bee.
- Implement Finance / Accounting Bee.
- Implement Inventory Bee.
- Introduce Advisor Bee.
- Implement cross-functional correlation.

Phase 3 — Governance & Controlled Actions

- Implement Guard Bee.
- Introduce approval workflows.
- Define business rules.
- Integrate selected business actions.
- Expand Audit Bee capabilities.

Phase 4 — Intelligent Business Operations

- Expand specialist agents.
- Introduce proactive alerts.

- Enable recurring investigations.
 - Add predictive and scenario analysis.
 - Expand automated, policy-controlled actions.
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14. Success Measures

The platform can be evaluated using measurable business and operational KPIs, including:

Business Intelligence

- Time required to answer business questions.
- Number of manual reports required.
- Accuracy and completeness of identified contributing factors.
- Evidence coverage for recommendations.

Operational Efficiency

- Reduction in manual analysis time.
- Number of investigations completed automatically.
- Agent execution time.
- Percentage of workflows requiring human intervention.

Governance

- Percentage of actions validated by Guard Bee.
- Number of unauthorised actions prevented.
- Audit completeness.
- Approval compliance.

Business Outcomes

Depending on the use case, organisations can measure changes in:

- Revenue.
- Gross margin.
- Lead conversion.
- Inventory availability.
- Stock-out frequency.

- Customer retention.
 - Operating costs.
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15. Future Vision

Following successful validation of the PoC, the solution could be expanded to include additional specialist agents such as:

- **Marketing Bee** — campaigns and customer acquisition.
- **Operations Bee** — operational efficiency.
- **Supplier Bee** — supplier performance and procurement.
- **HR Bee** — workforce-related business indicators.
- **Forecast Bee** — forecasting and scenario analysis.
- **Risk Bee** — business risk monitoring.

The platform could also evolve from reactive question answering into proactive monitoring.

Instead of waiting for an SME owner to ask:

“Why did my sales decrease?”

Queen Bee could proactively notify the owner:

“Sales have declined beyond the normal range this week. Inventory availability and customer conversion are potential contributing factors. Would you like to investigate?”

This would transform the system from a **business question-answering assistant** into a proactive **AI-powered SME performance monitoring platform**.

16. Conclusion

This proposal presents a **secure, AI-powered multilingual multi-agent assistant for SME business performance monitoring**, implemented as a serverless proof of concept.

The proposed Queen Bee architecture combines:

Natural-language interaction + Multilingual AI + Specialist Agents + Cross-functional Analysis + Security + Governance + Auditability

The central concept is to allow an SME owner to ask a business question naturally and have a coordinated team of AI agents investigate the relevant areas of the business.

The system does not simply provide an answer. It aims to show:

What happened → Why it may have happened → What evidence supports the finding → What can be done → Whether approval is required → What the AI agents did

By implementing this architecture as a serverless PoC, the project can demonstrate the feasibility of a secure and scalable approach to AI-assisted SME business performance monitoring while keeping the initial implementation focused and manageable.